LECTURE 3

3. Separable polynomials and separable extensions

Definition 3.1. We say that a root a (in a splitting field) of a polynomial f has *multiplicity* k if $f = (t-a)^k g$ with $(t-a) \nmid g$ (so that $g(a) \neq 0$).

Definition 3.2. An irreducible polynomial $f \in K[t]$ is *separable* if f has no multiple roots in a splitting field: $f = k(t-a_1)\cdots(t-a_n)$ for $k \in K$ with all a_i distinct. An irreducible polynomial that is not separable is called *inseparable*.

Example 3.3. The irreducible polynomial $g \in \mathbb{Q}[x]$ where $g = x^2 + 1 = (x+i)(x-i)$ is separable. Examples of inseparable irreducible polynomials are not so easy to find, as we will see later.

Definition 3.4. The *(formal) derivative* of a polynomial $f = a_0 + a_1 t + a_2 t^2 + \cdots + a_n t^n$ over an arbitrary field K is defined to be $Df = a_1 + 2a_2 t + \cdots + n a_n t^{n-1}$ ("incidentally" the same as the conventional derivative if $K = \mathbb{R}$ or $K = \mathbb{C}$).

Example 3.5. For the polynomial $f = x^3 - 3x^2 + 2x$, the derivative is $Df = 3x^2 - 6x + 2$.

Using only the definition, one can prove the following properties by a direct calculation:
(1) $D(f+g) = Df + Dg$;
(2) $D(f \cdot g) = Df \cdot g + f \cdot Dg$;
(3) $D(t-a)^k = k(t-a)^{k-1}$ for any $k \in \mathbb{N}$.

Theorem 3.6. (Multiple roots) *A root a (in a splitting field) of a polynomial f has multiplicity ≥ 2 if and only if a is also a root of Df (in other words, if $(t-a) \mid Df$, i.e. $(t-a) \mid (f, Df)$).*

Proof. If $f = (t-a)^k g$ with $k \geq 2$, then $Df = k(t-a)^{k-1} g + (t-a)^k Dg$ which is obviously divisible by $t-a$ (since $k-1 \geq 1$, the first summand is, and since the second is too, the sum is as well).

Conversely, if $f = (t-a)g$ with $(t-a) \nmid g$ (that is, $g(a) \neq 0$), then $Df = g + (t-a)^k Dg$ which is not divisible by $t-a$, since the first summand is not and the second is. □

Example 3.7. For $h = x^4 + 2x^2 + 1$, we have $Dh = 4x^3 + 4x$. By the theorem, since $\pm i$ is also a root of Dh, the root $\pm i$ has multiplicity strictly greater than 1. Indeed, in this case $\pm i$ has multiplicity 2.

Corollary 3.8. *A polynomial $f \in K[t]$ has no multiple roots in a splitting field if and only if $(Df,\ f) = 1$.*

Example 3.9. For $g = x^2 + 1$, the derivative $Dg = 2x$ satisfies $(Dg, g) = 1$ (by application of Euclid's algorithm). Hence by the corollary, the polynomial g has no multiple roots.

Remark 3.10. Note that the property to have (or to have no) multiple roots is easily decided **within** $K[t]$, since $Df \in K[t]$ and the Euclidean algorithm can be used to compute $(Df,\ f)$ (although the roots themselves may not be in K, but only in a larger splitting field, and these roots may not be known).

Corollary 3.11. (Separability) *Every irreducible polynomial over a field of characteristic zero is separable. An irreducible polynomial over a field of characteristic $p \neq 0$ is inseparable if and only if it has the form $f = a_0 + a_p t^p + a_{2p} t^{2p} + \cdots + a_{kp} t^{kp}$ (i.e. $f = g(t^p)$ for some $g \in K[t]$).*

Proof. See Exercise Sheet 3. □

Remark 3.12. Note that in characteristic p the condition is necessary and sufficient only for irreducible polynomials. However over some fields of characteristic p the polynomials of the form $g(t^p)$ are never irreducible: this means that all irreducible polynomials are separable over such a field. For example, one can show that over finite fields all irreducible polynomials are separable. Indeed, one can show that for a finite field K of characteristic p the mapping $x \mapsto x^p$ is an isomorphism (see Exercise Sheet 2). It follows that every element $a \in K$ can be represented in the form $a = b^p$ for some $b \in K$. Then, if $f \in K[t]$ was inseparable and irreducible, then, by the corollary above (using also for example Exercise 2.1(a)),

$$f = a_0 + a_p t^p + a_{2p} t^{2p} + \cdots + a_{kp} t^{kp} = b_0^p + b_p^p t^p + b_{2p}^p t^{2p} + \cdots + b_{kp}^p t^{kp} = (b_0 + b_p t + b_{2p} t^2 + \cdots + b_{kp} t^k)^p$$

which is not irreducible, a contradiction.

Example 3.13. So, to produce an inseparable irreducible polynomial, we must have an infinite field of characteristic $p \neq 0$. Here is an example. Let $K = \mathbb{Z}_p(u)$ be a simple transcendental extension of $\mathbb{Z}_p$, the field of residues mod p, and let $f(t) = t^p - u$. By the corollary above, if we show that f is irreducible, then f is inseparable. Certainly in a splitting field $L \supseteq K$ we have a root a of f. Then $a^p = u$ and $f = t^p - u = t^p - a^p = (t - a)^p$. So a is the only root of f of multiplicity p. Let us show that f is irreducible over K: suppose the opposite, that $f = gh$ for $g, h \in K[t]$. By the uniqueness of the decomposition, we must have $g = (t - a)^s$ for some $s \leq p - 1$. Since $g \in K[t]$, the constant coefficient of g, which is $\pm a^s$, must be in K, i.e. $a^s \in K$. However $(p, s) = 1$, hence there are $x, y \in \mathbb{Z}$ such that $xs + yp = 1$. Then $a = a^{xs+yp} = (a^s)^x (a^p)^y \in K$. This is however not true: if $a \in K$ then $a = c(u)/d(u)$ for some polynomials $c, d \in \mathbb{Z}_p[u]$ and $(c(u)/d(u))^p = u$ implies that $c(u)^p = d(u)^p u$, which is impossible, since the degree on the left is divisible by p, while on the right it is not.

Definition 3.14. An arbitrary (not necessarily irreducible) polynomial is *separable*, if all of its irreducible factors are separable.

Example 3.15. The polynomial $f = x^4 - 3x^3 + 3x^2 - 3x + 2 \in \mathbb{Q}[x]$ is the product $(x^2 + 1)(x - 1)(x - 2)$. Hence f is separable.

Of course every polynomial is separable over a field of characteristic 0.

Definition 3.16. An algebraic element $a \in L \supseteq K$ over K is *separable* if the minimal polynomial of a is separable. The algebraic extension $L : K$ is *separable* if every element of L is separable over K.

Note that every element in K is separable over K.

Example 3.17. As we saw, in characteristic 0 every (algebraic) extension is separable, and every algebraic extension of a finite field K is separable over K.

Lemma 3.18. (Separable-intermediate) *Suppose that $M \supseteq L \supseteq K$ are fields, and $M : K$ is separable. Then both $M : L$ and $L : K$ are separable too.*

Proof. For $L : K$ this is clear: the minimal polynomial for an element of L over K is the same separable one. Let $m \in M$ and let $g \in L[t]$ be the minimal polynomial of m over L. We need

to show that g is separable, i.e. has no multiple roots. Let f be the minimal polynomial of m over K. By the hypothesis f has no multiple roots. However $g \mid f$, since g is the minimal polynomial of m and $f(m) = 0$ too and both f and g are over L. Hence $f = gh$ for some $h \in L[t]$ and hence g has no multiple roots (for otherwise f would have). □

Example 3.19. As $\mathbb{Q}(\sqrt{2}, i) : \mathbb{Q}$ is separable, $\mathbb{Q}(\sqrt{2}, i) : \mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{2}) : \mathbb{Q}$ are too.

4. K-Automorphisms, Galois groups and the Galois correspondence

Definition 4.1. An *automorphism* of a field L is a bijection of L that preserves all field operations of L. The set S_L of all automorphisms of L can be shown to form a group $\operatorname{Aut} L$ with respect to composition of mappings.

Example 4.2. The map $\theta : \mathbb{Q}(\sqrt{2}) \to \mathbb{Q}(\sqrt{2})$ given by $a+b\sqrt{2} \mapsto a-b\sqrt{2}$ is an automorphism of $\mathbb{Q}(\sqrt{2})$.

Definition 4.3. Let $L \supseteq K$ be fields. An automorphism φ of L is a *K-automorphism* if $\varphi|_K = \varepsilon_K$. In other words, K-automorphisms are simply K-monomorphisms of L onto itself.

Example 4.4. The automorphism $\theta : \mathbb{Q}(\sqrt{3}) \to \mathbb{Q}(\sqrt{3})$ given by $a + b\sqrt{3} \mapsto a - b\sqrt{3}$ is a $\mathbb{Q}$-automorphism of $\mathbb{Q}(\sqrt{3})$.

Theorem 4.5. (Definition) *Let $L : K$ be a field extension. The set of all K-automorphisms of L forms a group $G = \operatorname{Gal}(L : K)$ with respect to composition of mappings, the* **Galois group** *of $L : K$.*

Proof. Since S_L, the set of all bijections of L, is a group, we only need to show that G is non-empty and closed under the group operation and under taking inverses. Clearly G is non-empty as $\varepsilon_L \in G$. Next we show that $ab \in G$ if $a, b \in G$. We know that ab is a bijection of L being a composition of bijections. A routine check shows that ab preserves all operations in L (Exercise Sheet 4). Finally $ab|_K = \varepsilon_K$ since $(ab)(k) = a(b(k)) = a(k) = k$ for any $k \in K$, since $a|_K = b|_K = \varepsilon_K$. Now we show that $a^{-1} \in G$ if $a \in G$. Again, we know that a^{-1} is a bijection of L. Then a routine check shows that a^{-1} preserves all operations in L: for example,

$$a^{-1}(\ell_1\ell_2) = a^{-1}(\ell_1)a^{-1}(\ell_2) \quad \Longleftrightarrow \quad a(a^{-1}(\ell_1\ell_2)) = a(a^{-1}(\ell_1)a^{-1}(\ell_2))$$

since a is a bijection. On the left we have $\ell_1\ell_2$, and on the right

$$a(a^{-1}(\ell_1)a^{-1}(\ell_2)) = a(a^{-1}(\ell_1)) \cdot a(a^{-1}(\ell_2)) = \ell_1\ell_2,$$

the same as the LHS, as required. Finally $a^{-1}|_K = \varepsilon_K$ since $a^{-1}(k) = k$ for any $k \in K$. Indeed, since a is a bijection, we have

$$a(a^{-1}(k)) = a(k) \quad \Longleftrightarrow \quad k = a(k),$$

which is true since $a|_K = \varepsilon_K$. □

We proved the following as part of the proof for Theorem 2.11 (finite normal=splitting):

Lemma 4.6. (Basic lemma on K-monomorphisms) *Suppose that $K \subseteq L \subseteq M$ are fields and L is a splitting field for $f \in K[t]$. Then any K-automorphism φ of M permutes the roots of f and leaves L invariant: $f(a) = 0 \Rightarrow f(\varphi(a)) = 0$ and $\varphi(L) = L$.*

One can actually see from the proof, that, in fact $f(a) = 0 \Rightarrow f(\varphi(a)) = 0$ for any $f \in K[t]$ and any K-monomorphism φ (even if the splitting field is beyond the domain or range of φ), and that always $\varphi(K(u_1, \ldots, u_s)) = K(\varphi(u_1), \ldots, \varphi(u_s))$, where $\{u_1, \ldots, u_s\}$ is some subset of roots of f.

Remark 4.7. We can now discuss the main idea of Galois Theory. If $L : K$ is a finite normal extension then L is a splitting field of some $f \in K[t]$, i.e. $L = K(a_1, \ldots, a_m)$, where $\{a_1, \ldots, a_m\}$ are all the distinct roots of f. The elements of $G = \mathrm{Gal}(L : K)$ then permute the a_i's, and clearly, the action of $g \in G$ on L is completely determined by its action on the set $\{a_1, \ldots, a_m\}$. In other words, the mapping

$$g \to \begin{pmatrix} a_1 & a_2 & \cdots & a_m \\ g(a_1) & g(a_2) & \cdots & g(a_m) \end{pmatrix}$$

is an isomorphism of G onto a subgroup of S_m. This is a finite group, hence its properties and its structure are somehow easier to understand than those of the (in general infinite) field L. Of course, we already had a finite invariant $[L : K]$, the dimension of L as a vector space over K, but this is too crude an instrument, so to speak; e.g. we cannot even decide, whether there are finitely or infinitely many intermediate fields between L and K (because there are, of course, infinitely many subspaces in general). However the Galois group will give us important information on the structure of the extension, including the number of intermediate fields.

Example 4.8. 1. Let G be the Galois group of $\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}$. Then $G = \{e\}$ consists of only the identity mapping of $\mathbb{Q}(\sqrt[3]{2})$. Indeed, if $g \in G$, then $g(\sqrt[3]{2})$ must also be a root of $t^3 - 2 \in \mathbb{Q}[t]$, by the basic lemma on K-monomorphisms. However there are no roots of $t^3 - 2$ in $\mathbb{Q}(\sqrt[3]{2})$ other than $\sqrt[3]{2}$. Hence $g(\sqrt[3]{2}) = \sqrt[3]{2}$ and hence $g|_{\mathbb{Q}(\sqrt[3]{2})} = \varepsilon_{\mathbb{Q}(\sqrt[3]{2})}$ for all $\mathbb{Q}$-automorphisms g. This happens because the extension is not normal; it does not contain all roots of $t^3 - 2$.

2. Let G be the Galois group of $\mathbb{Q}(i, \sqrt{2}) : \mathbb{Q}$. Since i is a root of $t^2 + 1 \in \mathbb{Q}[t]$, every $g \in G$ must send i to a root of $t^2 + 1$, to $\pm i$; again by the basic lemma on K-monomorphisms. Similarly $\sqrt{2}$ is a root of $t^2 - 2$, so that $g(\sqrt{2})$ can only be $\pm\sqrt{2}$, and g is determined by $g(i)$ and $g(\sqrt{2})$. In fact all four possibilities really occur, that is, the group G has all these 4 elements. To see this, first consider $\mathbb{Q}(i, \sqrt{2})$ as a simple extension of $\mathbb{Q}(\sqrt{2})$. By the theorem on isomorphisms of simple extensions, there is an automorphism g_1 such that $g_1(i) = -i$ and $g_1|_{\mathbb{Q}(\sqrt{2})} = \varepsilon_{\mathbb{Q}(\sqrt{2})}$, i.e. $g_1(\sqrt{2}) = \sqrt{2}$. We can also write explicitly g_1 as a mapping of $\mathbb{Q}(i, \sqrt{2})$ (which has basis $1,\ \sqrt{2},\ i,\ i\sqrt{2}$ over $\mathbb{Q}$): $g_1(a + b\sqrt{2} + ci + di\sqrt{2}) = a + b\sqrt{2} - ci - di\sqrt{2}$.

Next, consider $\mathbb{Q}(i, \sqrt{2})$ as a simple extension of $\mathbb{Q}(i)$. Again, there is $g_2 \in G$ such that $g_2(\sqrt{2}) = -\sqrt{2}$ and $g_2(i) = i$. We have $g_2(a + b\sqrt{2} + ci + di\sqrt{2}) = a - b\sqrt{2} + ci - di\sqrt{2}$.

By the Theorem-Definition, we know that $g_3 = g_1 g_2 \in G$ too; we have $g_3(i) = -i$ and $g_3(\sqrt{2}) = -\sqrt{2}$. The fourth element is e, the identity mapping. It can be easily computed that $g_i^2 = e$ for all $i = 1, 2, 3$, that G is commutative, and that $g_1 g_2 = g_3$, $g_2 g_3 = g_1$, $g_3 g_1 = g_2$. Hence $G \cong C_2 \times C_2$. (Alternatively, instead of appealing to the commutative aspect, one can refer to $|G| \leq 4$ to justify that $G \cong C_2 \times C_2$.)

Definition 4.9. Let G be the Galois group of an extension $L : K$. For a subgroup $H \leq G$, the *fixed field of* H is $H^\dagger = \{x \in L \mid h(x) = x \text{ for all } h \in H\}$. Alternative notation: L^H

That this is really a subfield is a routine check based on the definitions.

Definition 4.10. Let G be the Galois group of an extension $L : K$. For an intermediate field $K \subseteq V \subseteq L$, the *subgroup of V-automorphisms* is $V^* = \{g \in G \mid g|_V = \varepsilon_V\}$. Alternative notation: $\mathrm{Aut}(L : V)$.

Again, the proof that it is a subgroup is a routine check. Note that V^* is also by definition the Galois group of $L : V$.

The mapping †, from the set of all subgroups of G to the set of all intermediate fields, and the reverse $*$, are known as the **Galois Correspondence**. Our goal is the Fundamental Theorem of Galois Theory which states that if $L : K$ is finite, normal and separable, then † and $*$ are bijections and inverses of one another.

The following properties follow easily from the definitions:

(1) if $H_1 \leq H_2$, then $H_1^\dagger \supseteq H_2^\dagger$;
(2) if $V_1 \subseteq V_2$, then $V_1^* \geq V_2^*$;
(3) $\{e\}^\dagger = L$;
(4) $K^* = G$.

For example, let us prove (1): if $\ell \in H_2^\dagger$ then $h(\ell) = \ell$ for all $h \in H_2$, and certainly for all $h \in H_1$ since $H_1 \leq H_2$ by hypothesis. Hence $\ell \in H_1^\dagger$ by definition.

Example 4.11. Let us consider the Galois group G of $\mathbb{Q}(i, \sqrt{2}) : \mathbb{Q}$ as in Example 4.8(2). One can see that the only proper subgroups of G are $H_1 = \{e, g_1\}$, $H_2 = \{e, g_2\}$ and $H_3 = \{e, g_3\}$. Let us find $H_1^\dagger$. Clearly $\ell \in H_1^\dagger \Leftrightarrow g_1(\ell) = \ell$. Write $\ell = a + b\sqrt{2} + ci + di\sqrt{2}$. Then

$$g_1(\ell) = a + b\sqrt{2} - ci - di\sqrt{2} = \ell \quad \Longleftrightarrow \quad c = d = 0$$

(since $1,\ \sqrt{2},\ i,\ i\sqrt{2}$ is a basis over $\mathbb{Q}$). Thus $H_1^\dagger = \mathbb{Q}(\sqrt{2})$. A similar calculation shows that $H_2^\dagger = \mathbb{Q}(i)$. Find $H_3^\dagger$ as an exercise (Exercise Sheet 5).

Let us find $\mathbb{Q}(\sqrt{2})^*$: clearly $e \in \mathbb{Q}(\sqrt{2})^*$ and $g_1 \in \mathbb{Q}(\sqrt{2})^*$, but $g_2, g_3 \notin \mathbb{Q}(\sqrt{2})^*$. Thus $\mathbb{Q}(\sqrt{2})^* = H_1$.

Note that here, in accordance with the Fundamental Theorem, we have indeed $(H_1^\dagger)^* = H_1$ and $(\mathbb{Q}(\sqrt{2})^*)^\dagger = \mathbb{Q}(\sqrt{2})$, as it should be. (Similar calculations give $(H_i^\dagger)^* = H_i$ for all $i = 1, 2, 3$.)

As a substantial part of the proof of the Fundamental Theorem, we will prove the following:

Theorem 4.12. (Degree over the fixed field) *Suppose that G is a finite subgroup of the group of automorphisms of a field L. Then $|G| = [L : G^\dagger]$, where $G^\dagger$ is the fixed field of G.*